

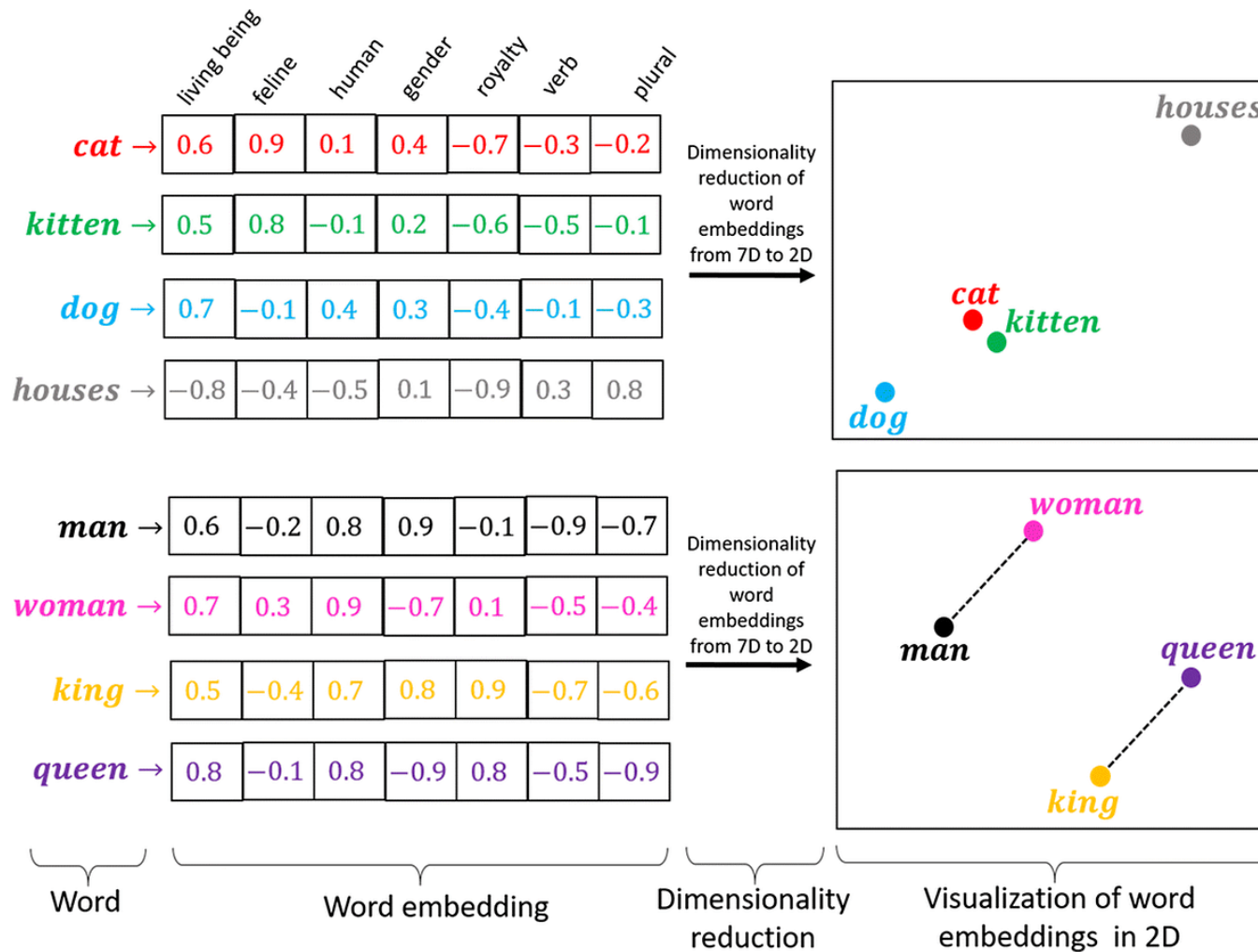
Procesamiento del lenguaje natural (NLP) II

Francisco Jáñez Martino

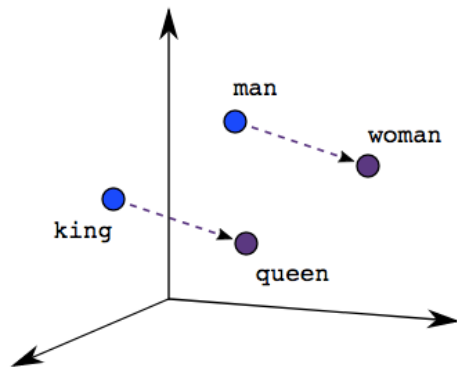
Módulo 3

Aprendizaje Automático Aplicado

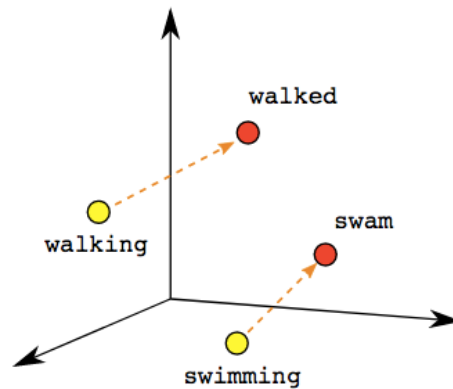
Word Embeddings



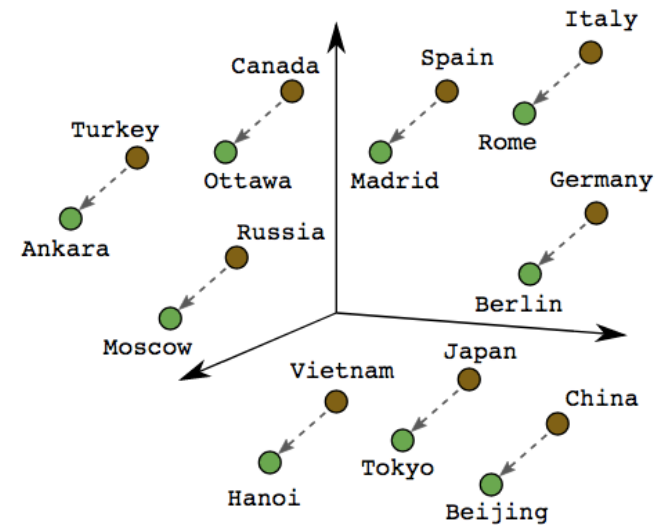
Word Embeddings



Male-Female

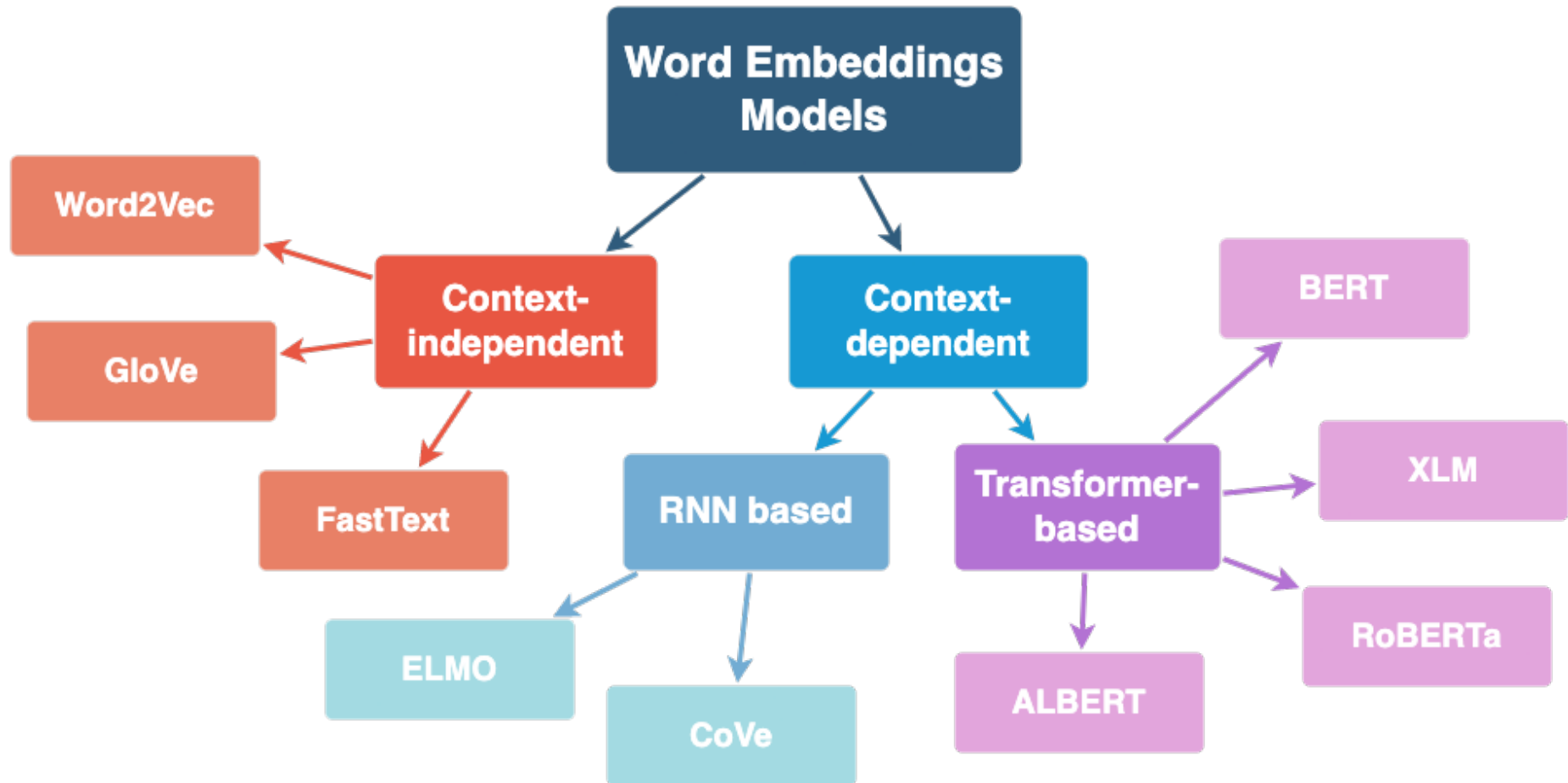


Verb Tense

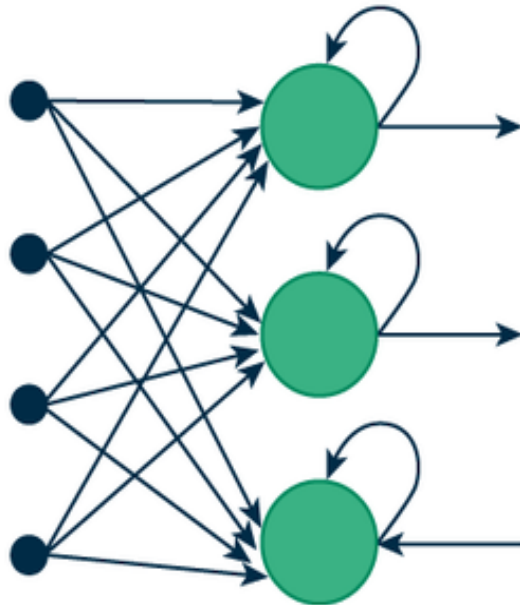


Country-Capital

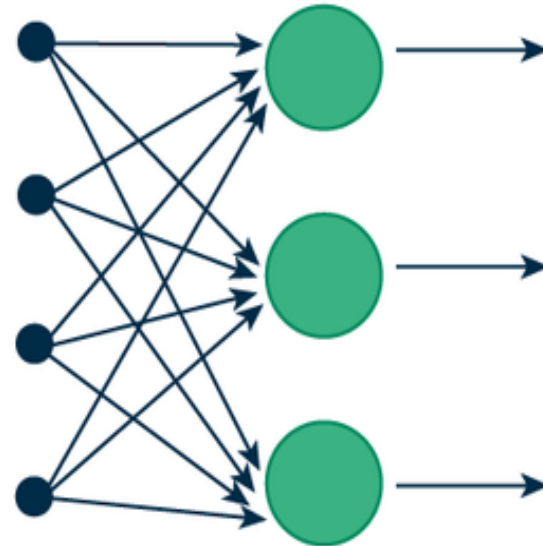
Word Embeddings



Recurrent Neural Networks (RNNs)

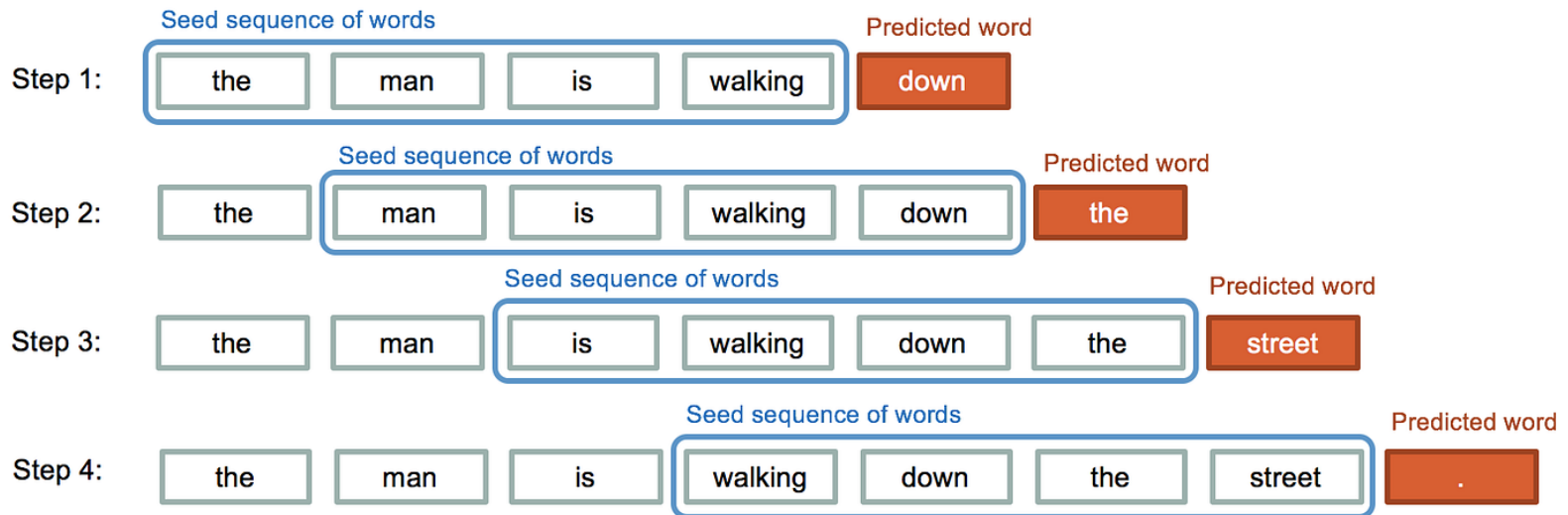


(a) Recurrent Neural Network

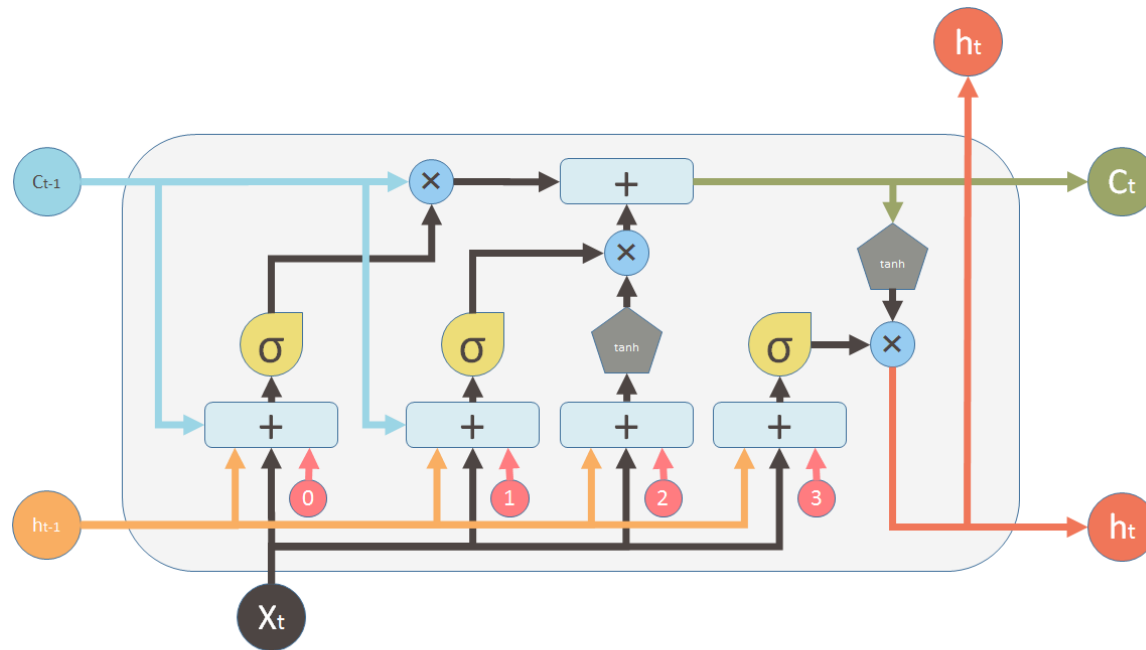


(b) Feed-Forward Neural Network

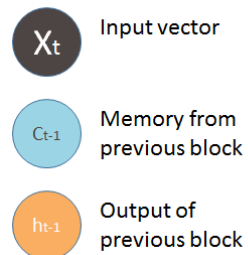
Recurrent Neural Networks (RNNs)



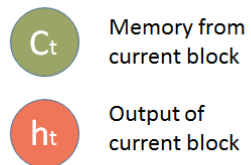
LSTM



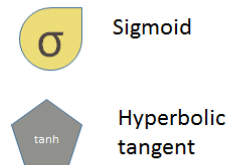
Inputs:



outputs:

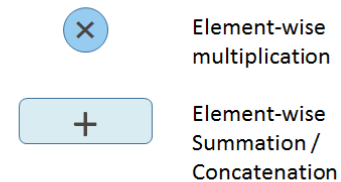


Nonlinearities:



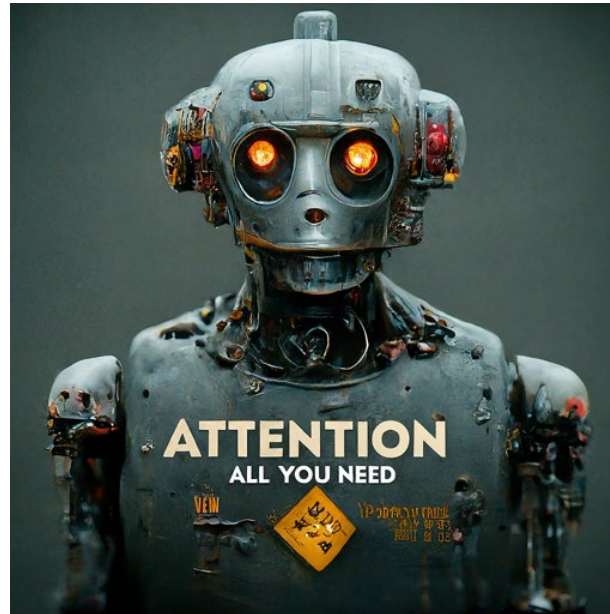
Bias: 0

Vector operations:



Modelos de atención y Transformers

Los **modelos de atención** mecanismos que permiten a los modelos de aprendizaje profundo enfocar su procesamiento en partes relevantes de una secuencia de datos, optimizando la interpretación y generación de información.



Los **Transformers** son un tipo de modelo de IA basado en redes neuronales, diseñado para procesar secuencias de datos mediante mecanismos de atención, fundamentales en modelos de lenguaje como GPT o BERT

Modelos de atención y Transformers

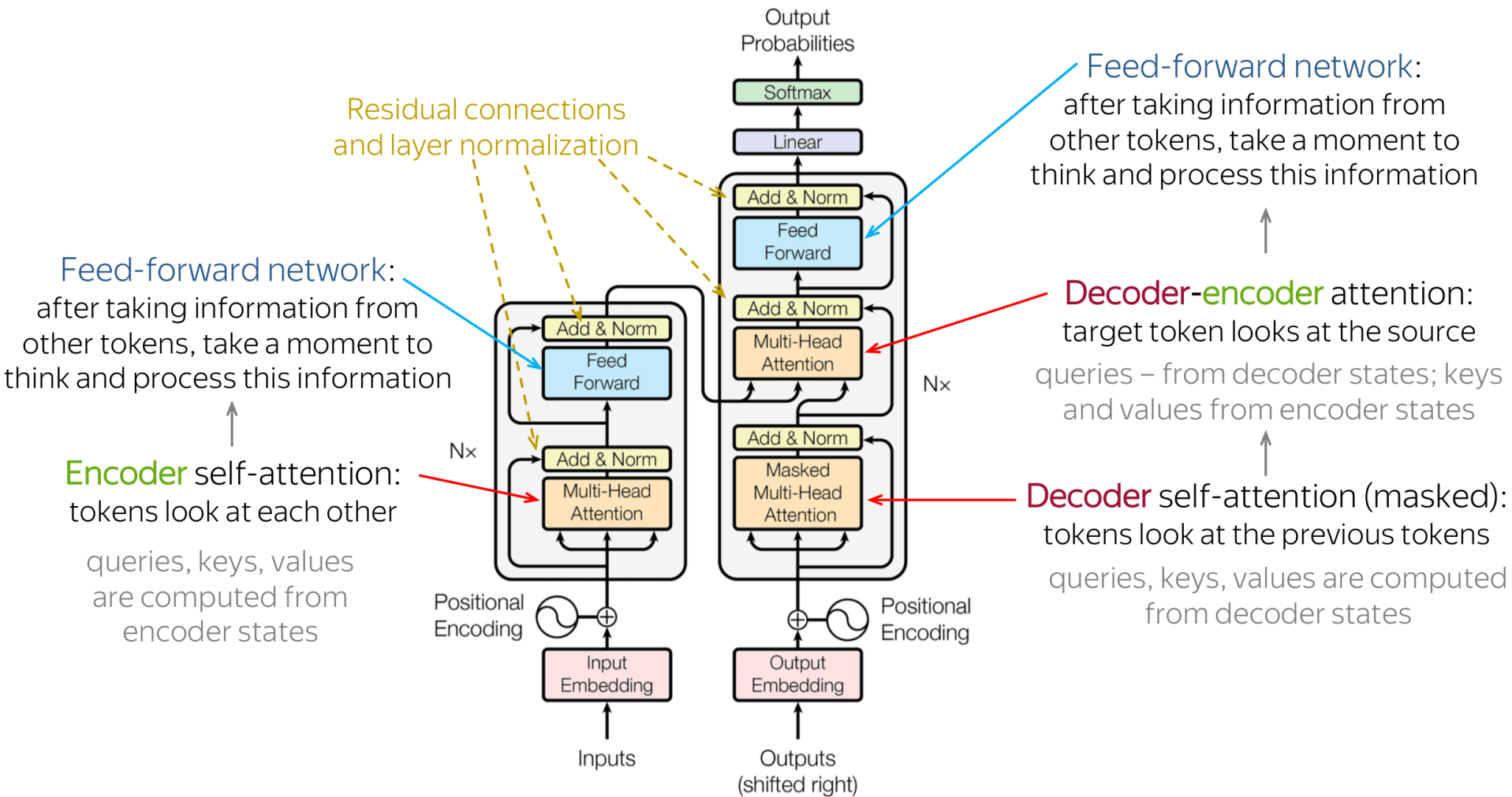
- **Consultas (Queries, Q):** Representan los elementos que buscan información.
- **Claves (Keys, K):** Actúan como identificadores para recuperar información relevante.
- **Valores (Values, V):** Contienen la información asociada a cada clave.

$$\text{Attention}(q, k, v) = \overbrace{\text{softmax}\left(\frac{qk^T}{\sqrt{d_k}}\right)}^{\text{Attention weights}} v$$

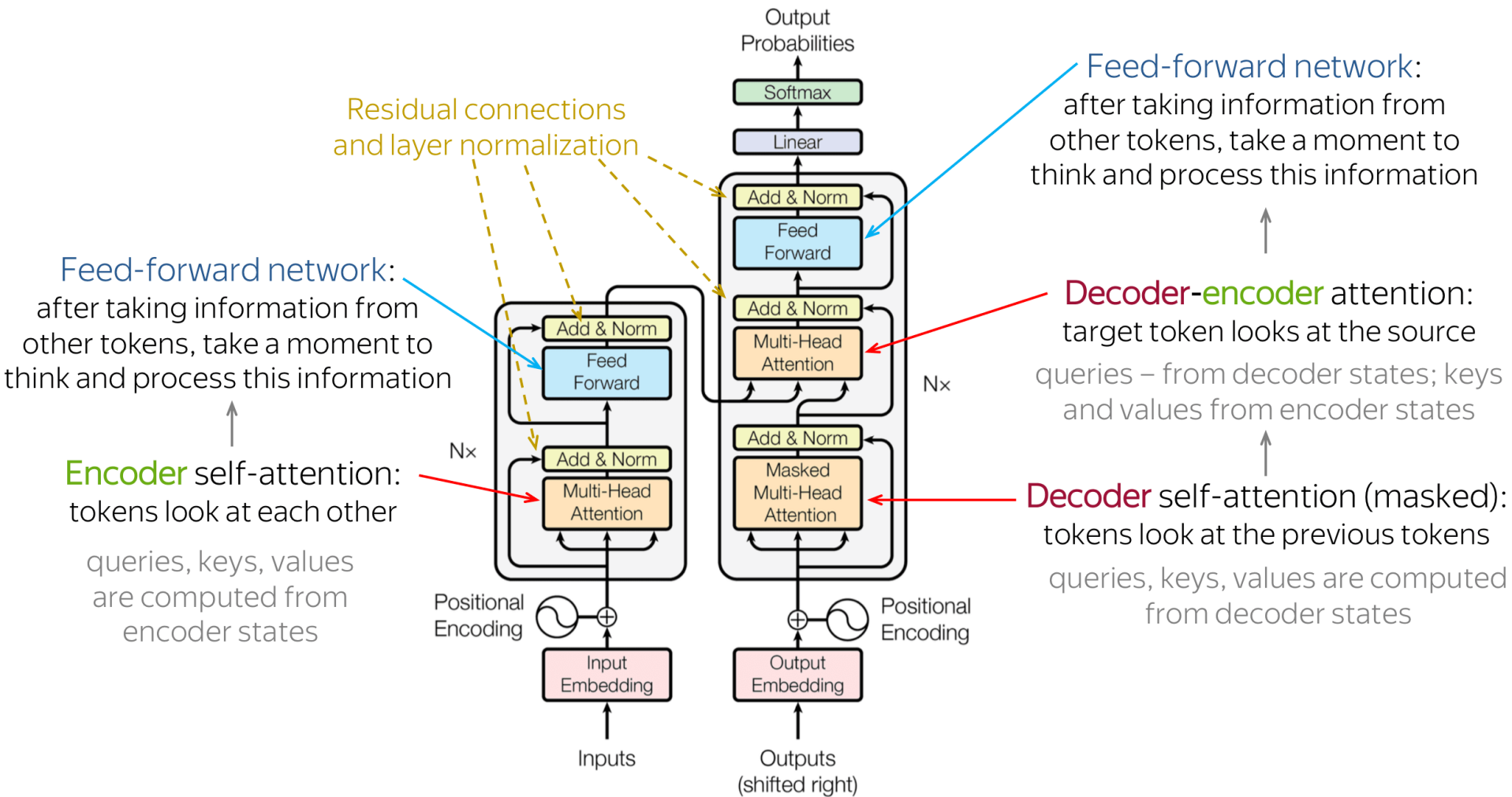
from to

vector dimensionality of K, V

Modelos de atención y Transformers

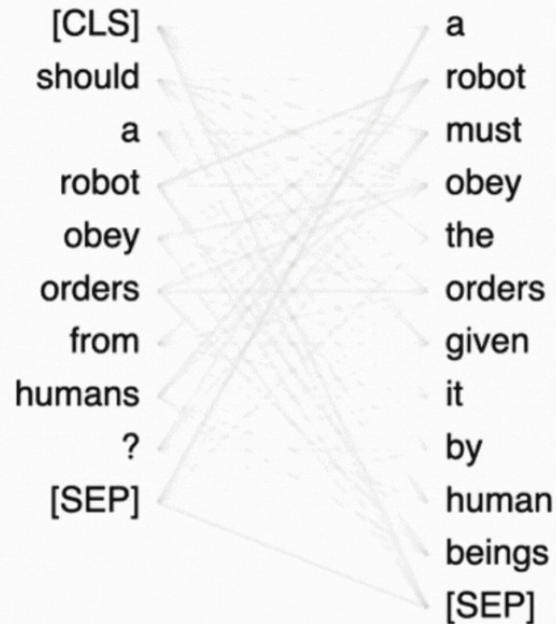


Modelos de atención y Transformers

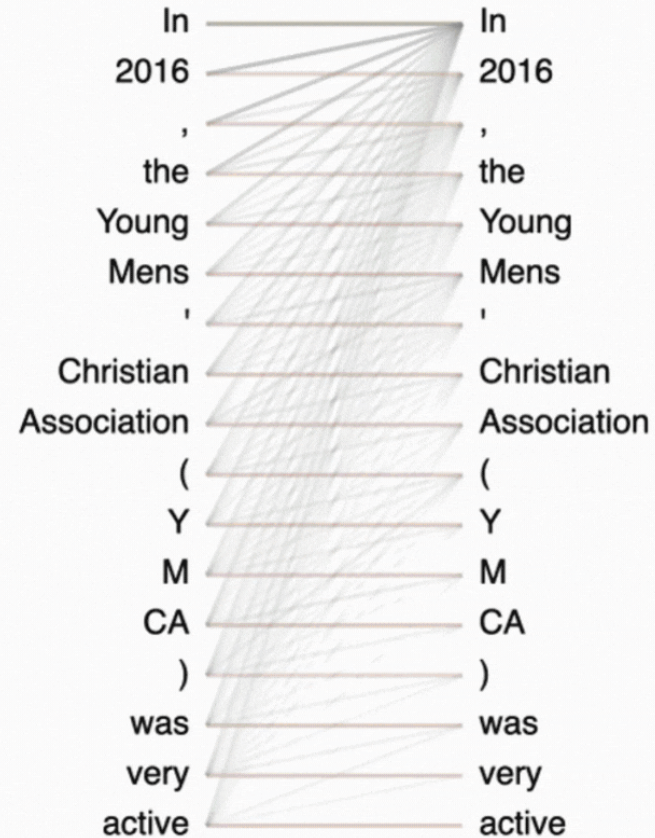


Modelos de atención y Transformers

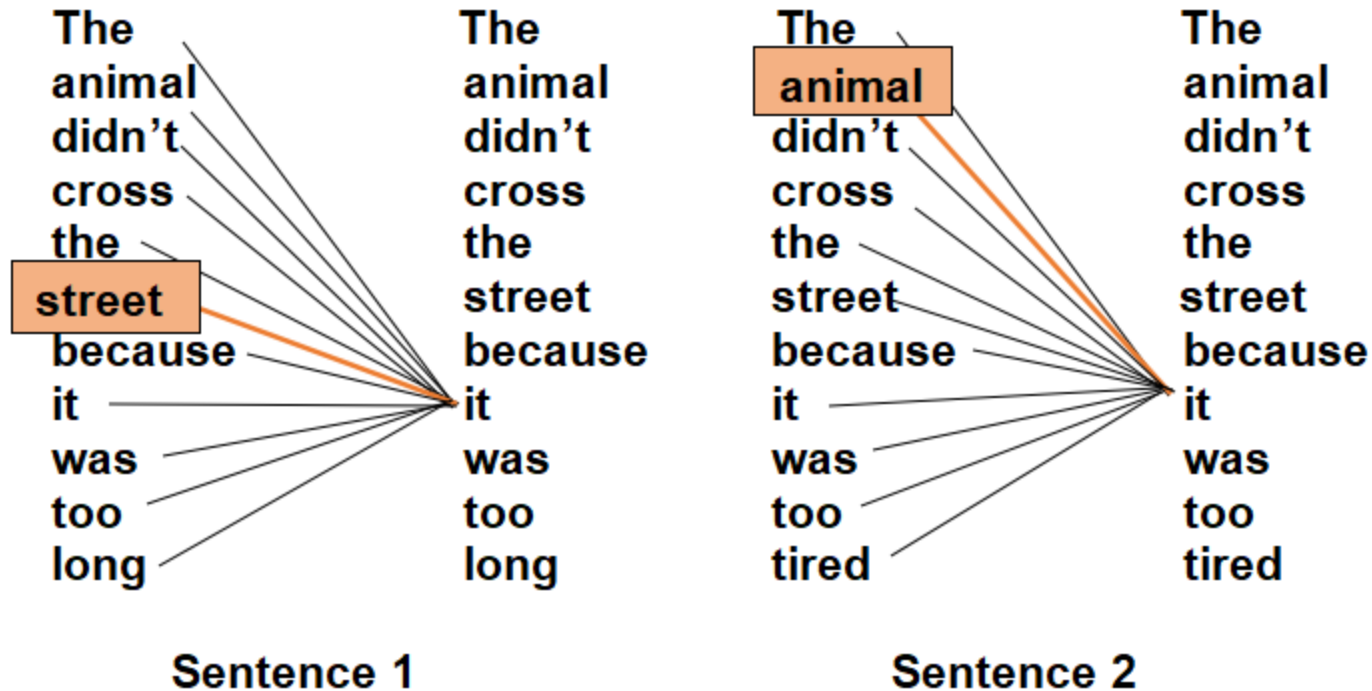
Layer: 0 ▾ Attention: Sentence A -> Sentence B ▾



Layer: 0 ▾ Attention: All ▾



Modelos de atención y Transformers



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